

Yushi Ye

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Research Interests

My research interests focus on developing efficient and reliable large language models, particularly diffusion large language models (dLLMs), and studying their applications in multimodal learning.

Education

Shanghai Jiao Tong University 2024.9-2027.3 (Expected)
M.E in Information Engineering

Shanghai Jiao Tong University 2020.9-2024.6
B.E. in Information Engineering (IEEE Honor Class)
B.S in Mathematics and Applied Mathematics (Dual Bachelor's Degree Program)

Research Experience

Shanghai Jiao Tong University 2024.9-2027.3 (Expected)
Master's Student in Information Engineering
Advisors: Assoc. Prof. Jiangchao Yao, Prof. Yanfeng Wang

- Conducted research on diffusion large language models, efficient inference, and multimodal learning.

Zhejiang Tmall Technology Co., Ltd. 2025.11-2026.7
Research Intern, Taotian Group — Pailitao Algorithm
Mentor: Xu Chen

- Conducted research on efficient diffusion LLM inference, leading to Ripple-Pivot Search (RPS), a training-free active parallel decoding method.

Publications

Ripple-Pivot Search: Active Parallel Decoding for Diffusion Large Language Models arXiv Preprint [🔗](#)

Yushi Ye, Xu Chen, Haoyun Jiang, Jinsong Lan, Xiaofeng Cao,
Xiangtao Li, Ivor Tsang, Yanfeng Wang, Tanghaihong, Jiangchao Yao

- Proposed **Ripple-Pivot Search (RPS)**, a training-free decoding method that uses lookahead evaluation to identify pivot tokens whose commitment reduces uncertainty across remaining masked positions, accelerating diffusion LLM inference while preserving generation quality.

Rejection Mixing: Fast Semantic Propagation of Mask Tokens for Efficient DLLM Inference CVPR Poster 2026 [🔗](#)

Yushi Ye*, Feng Hong*, Huangjie Zheng, Xu Chen, Zhiyong Chen, Yanfeng Wang, Jiangchao Yao

- Developed **ReMix**, a training-free inference framework for Diffusion LLMs (DLLMs) that resolves the "combinatorial contradiction" in parallel decoding. By introducing a Continuous Mixing State and a rejection-based refinement rule, ReMix enables tokens to resolve semantic conflicts in a continuous space before discretization.

Wide-In, Narrow-Out: Revokable Decoding for Efficient and Effective

ICLR Poster 2026 [🔗](#)

DLLMs

Feng Hong*, Geng Yu*, **Yushi Ye**, Haicheng Huang, Huangjie Zheng, Ya Zhang, Yanfeng Wang, Jiangchao Yao

- Developed **WINO**, a revokable parallel decoding algorithm designed to address quality degradation in diffusion-based LLMs under high-parallel settings. By integrating a self-verification mechanism, WINO effectively mitigates error accumulation, achieving a dual breakthrough in both generation fidelity and inference throughput.

Learning to Instruct for Visual Instruction Tuning

NeurIPS Poster 2025 [🔗](#)

Zhihan Zhou*, Feng Hong*, Jiaan Luo, **Yushi Ye**,

Jiangchao Yao, Dongsheng Li, Bo Han, Ya Zhang, Yanfeng Wang

- Developed **LIT**, a novel training algorithm designed to mitigate overfitting and shortcut learning in Visual Instruction Tuning (VIT). By guiding the model to generate accurate image-grounded instructions alongside its responses, LIT significantly enhances multimodal understanding and generation capabilities with zero additional data and negligible computational overhead.

Uncover the balanced geometry in long-tailed contrastive language-image pretraining

Mach Learn Journal 2025 [🔗](#)

Zhihan Zhou, **Yushi Ye**, Feng Hong, Peisen Zhao, Jiangchao Yao, Ya Zhang, Qi Tian, Yanfeng Wang

- Developed **GeoCLIP**, an enhanced CLIP pre-training framework for long-tailed distributions. By constructing pseudo-clusters and aligning geometrically balanced structures, our method achieves an optimal balance between uniformity and discriminativity in the semantic space, significantly boosting model robustness and generalization in long-tailed scenarios.

Others

English Proficiency

- TOEFL: 111/120

Honors & Awards

- Zhiyuan Honors Scholarship(2020)
- First-Class Graduate Freshman Academic Scholarship (2024)
- First-Class Graduate Comprehensive Academic Scholarship (2025)

Academic Service

Reviewer of ICML, NeurIPS, AAAI, TMLR